

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 16

14 July 2025

Overview:

In today's session, we were introduced with TensorFlow, a powerful and flexible open-source framework developed by Google for building and training machine learning and deep learning models. It is widely used by researchers and developers for tasks involving numerical computation and large-scale machine learning.

Learning Objectives:

- Understand what TensorFlow is and why it's used
- Learn the basic structure of a TensorFlow program
- Explore core components like tensors, operations, and graphs
- Get familiar with how TensorFlow supports model building, training, and deployment

What is TensorFlow?

TensorFlow is an open-source framework for machine learning and artificial intelligence that was developed by Google Brain. It was designed to facilitate the development of machine learning models, particularly deep learning models by providing tools to easily build, train and deploy them across different platforms.

TensorFlow supports a wide range of applications from natural language processing and computer vision to time series forecasting and reinforcement learning.

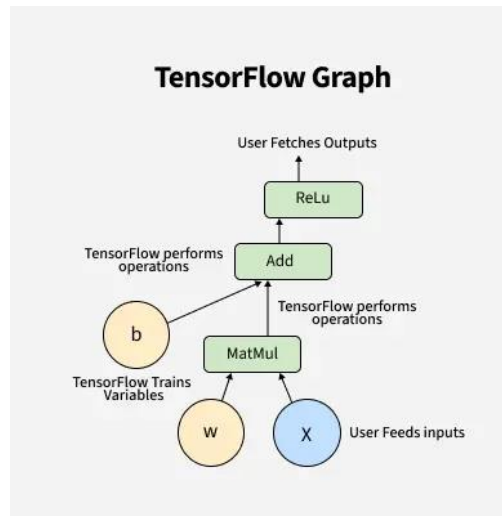
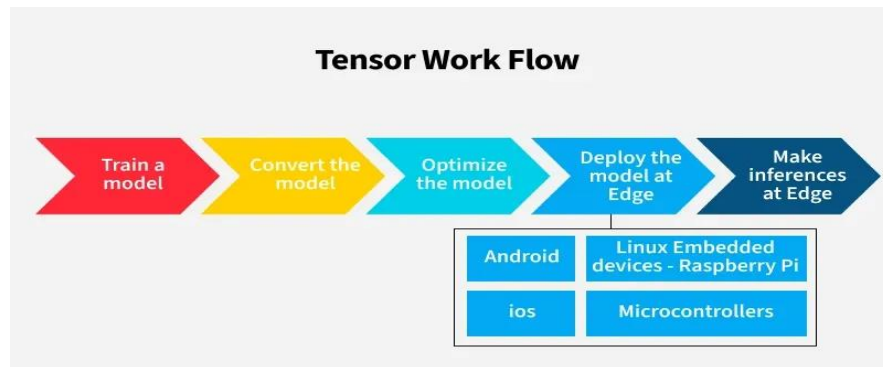


Fig. TensorFlow

Features of TensorFlow

- Scalability
- Comprehensive Ecosystem
- Automatic Differentiation (Auto grad)
- Multi-language Support
- TensorFlow Serving and TensorFlow Model Optimization

Working of TensorFlow



Step 1: Train a Model

Step 2: Convert the Model

Step 3: Optimize the Model

Step 4: Deploy the Model

Step 5: Make Inferences at the Edge

Install TensorFlow:

```
pip install tensorflow
```

Why TensorFlow?

TensorFlow is used because it provides a powerful, flexible, and efficient way to build and train machine learning and deep learning models. It was developed by Google to handle complex numerical computations and large-scale machine learning tasks.

Conclusion:

TensorFlow gave us our first taste of a professional-grade framework used in real-world AI. Its ability to simplify complex mathematical computations while remaining efficient and scalable makes it a top choice in the machine learning world. We look forward to building actual models using it in the coming sessions.